

Carry-Over Lottery Allocation: Practical Incentive-Compatible Drafts

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

Introduction

For decades, the NBA has navigated the tradeoff between helping weak teams via the draft and the tanking that ensues. Carry-Over Lottery Allocation (COLA) solves the dilemma. COLA defines weakness using multi-season playoff results, not single-season records. Thus, tanking need not be rewarded. Draft priority carries over between seasons, being reduced when teams have playoff success or make high draft picks. Over time, more help is directed to weaker teams.

COLA is a framework of mechanisms where each may have different rules on how draft priority changes. COLA's definition includes the NBA's own 3-2-1 lottery as an edge case: minimal carry-over exclusions and no reward for losing.

We prove that, under reasonable assumptions, COLA can be incentive-compatible while helping the weakest team, circumventing Banchio and Munro's SSAC 2020 impossibility result for record-based lotteries. Only one assumption requires justification: teams prefer winning games over manipulating the pool of lottery teams. Others are standard assumptions.

Methods

The incentive-compatibility proof is standard; novelty comes from the reframe that makes it possible.

We discuss tradeoffs within the COLA framework by identifying tunable dials:

- How does a team's draft priority increase?
- How does the priority decrease?
- How does priority map to lottery odds and draft positions?

ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move of realistic fictional leagues, lets us compare different draft mechanisms using these criteria:

- **Targeted help.** How much help do the highest priority teams receive?
- **Long-term parity.** Can all teams be competitive, as measured by long playoff droughts?
- **Incentive compatibility.** Does the mechanism reward losing?

Simplicity is assessed qualitatively. In the paper, we analyze the impact if model assumptions are violated, and how to mitigate issues, at a cost of complexity.

Results

We ran nine mechanisms on the same 30 simulated leagues for 25 seasons each. We compare mechanisms by paired differences with 95% confidence intervals.

The old NBA lottery rewards losing, causing tanking, which disqualified it. 3-2-1 removes that reward but does not help weak teams. Simple COLA also removes it and, unlike 3-2-1, directs significant help to weak teams, where weakness is drought length (Table 1). On parity, all three are statistically indistinguishable on the wait for a series win; on the wait to return to the playoffs, the NBA lotteries keep a slight edge (0.18 seasons against 3-2-1, interval 0.08 to 0.28). On simplicity, Simple COLA wins outright: no lottery is needed (Table 2). It is one example among many in the family that all solve the core dilemma. Full-lottery COLA, another, matches 3-2-1 on both waits while still helping more.

Rule	Incentive compatibility	Targeted help		Long-term parity	
	Help for a worse record places gained, 5th-worst to worst	Help for a longer wait places gained, 5th-longest to longest	Longest wait for a series win or top-3 pick seasons	Wait to return to the playoffs seasons, average	Wait for the next series win seasons, average
Old NBA lottery	+1.32	0	16.4	2.89	4.65
New lottery, 3-2-1	-0.63	0	16.3	2.94	4.47
Simple COLA	-0.94	6.3	5.5	3.13	4.58
Full-lottery COLA	0	1.6	11.9	2.89	4.51

Table 1. Four rules on the three criteria, on the same 30 simulated leagues of 25 seasons (means over leagues). Help counts the draft places one team gains when only its record, or only its wait, changes, so a positive number is an earlier pick. The full paper prices the reward for losing from every starting position, for all nine rules we ran.

Rule	How priority increases	How it decreases	How it maps to picks
Old NBA lottery	With a worse record this season	Nothing carries over	Top four drawn by record odds; 5 to 14 by record
New lottery, 3-2-1	With a worse record this season	No repeat #1 or third straight top-5	All 16 drawn by 2/3/2/1 balls; worst three pick by 12
Simple COLA	+1 each season without a series win	To zero after a series win or a top-3 pick	In priority order, no lottery; ties to most wins
Full-lottery COLA	+1 each season without a series win	Cut by a quarter to all, the deeper the run or the higher the pick	All 14 picks drawn, odds proportional to priority

Table 2. The four rules on the three dials above, one setting per row. Keeping a lottery night is one of the settings; the full paper turns every dial across nine rules.

Conclusion

COLA shows that helping the weakest teams and ending tanking are not mutually exclusive. It only looked that way because "weakest" was measured by record. Simple COLA resolves the dilemma without complexity, and the wider family lets a league choose its own balance of targeted help, parity, simplicity, and robustness.